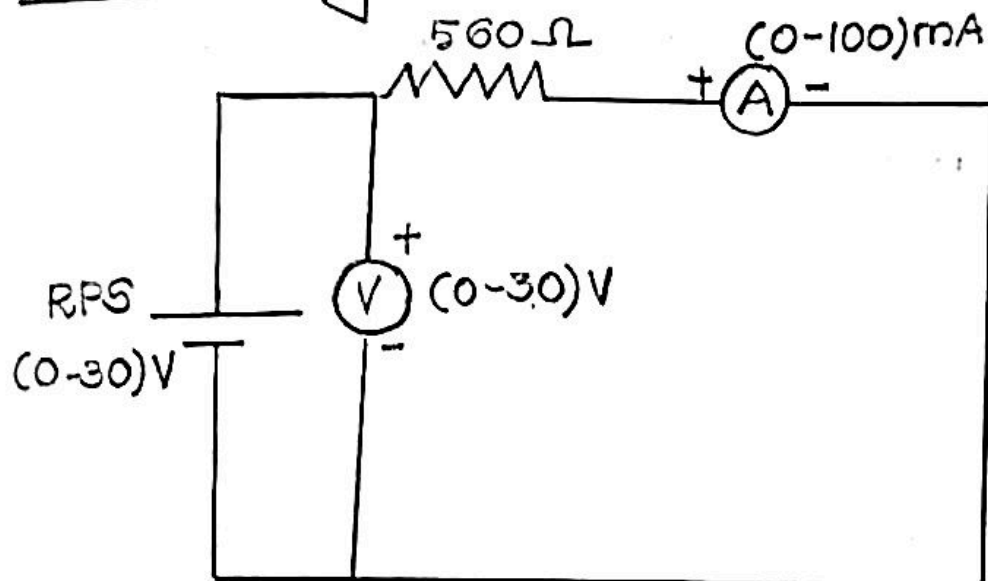


Circuit diagram:



Tabular Column:

S.No	Voltage (V)	Current (mA)	Experimental value ($R = V/I$) in $k\Omega$	Theoretical value of R (Ω)
1.	1	2.2	0.45	560
2.	2	2.8	0.71	560
3.	3	6.3	0.48	560
4.	4	8	0.5	560
5.	5	9.8	0.51	560

Avg: 0.53 $k\Omega$

Ex.No.1

Verification of Ohm's law & Kirchhoff's law1) a) Verification of Ohm's law.Aim:

To verify Ohm's law for a given resistive network

Apparatus Required:

Sl. No	Apparatus Name	Range	Quantity
1.	RPS	(0-30)V	1
2.	Ammeter	(0-100)mA	1
3.	Voltmeter	(0-30)V	1
4.	Resistor	560 Ω	1
5.	Bread. board & Connecting wires	-	Required

Theory:

Ohm's law is a fundamental principle in electricity that explains the relationship between, voltage, current and resistance in an electrical circuit.

It states that the current (I) flowing through a conductor is directly proportional to the voltage (V) applied across it and inversely proportional to the resistance (R) of the conductor, expressed mathematically as $V = I \times R$

Theoretical calculation:

$$V = IR$$

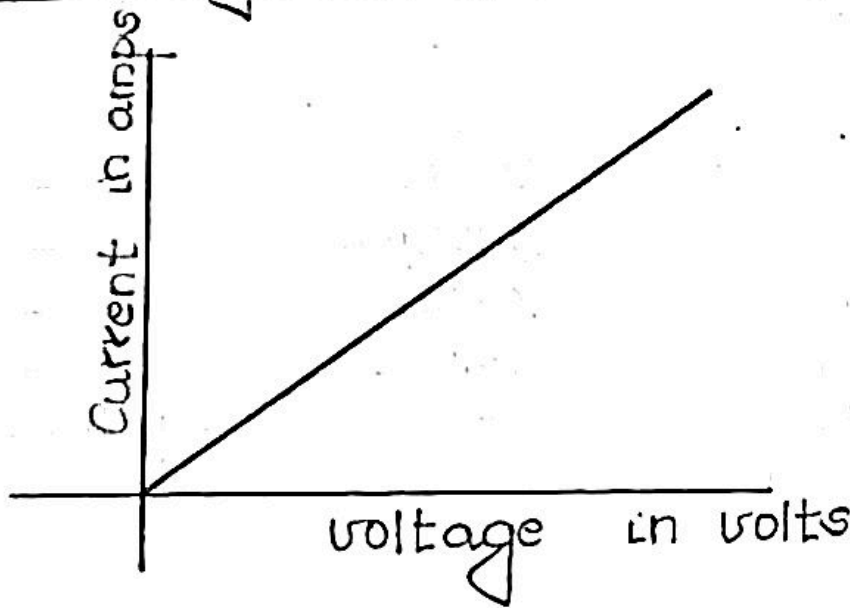
$$I = \frac{V}{R} \quad \text{or} \quad R = \frac{V}{I}$$

If $V = 1V$, current = $2.2mA$

$$R = \frac{1}{2.2 \times 10^{-3}} = 0.45 k\Omega$$

$$= 450\Omega$$

Model graph:



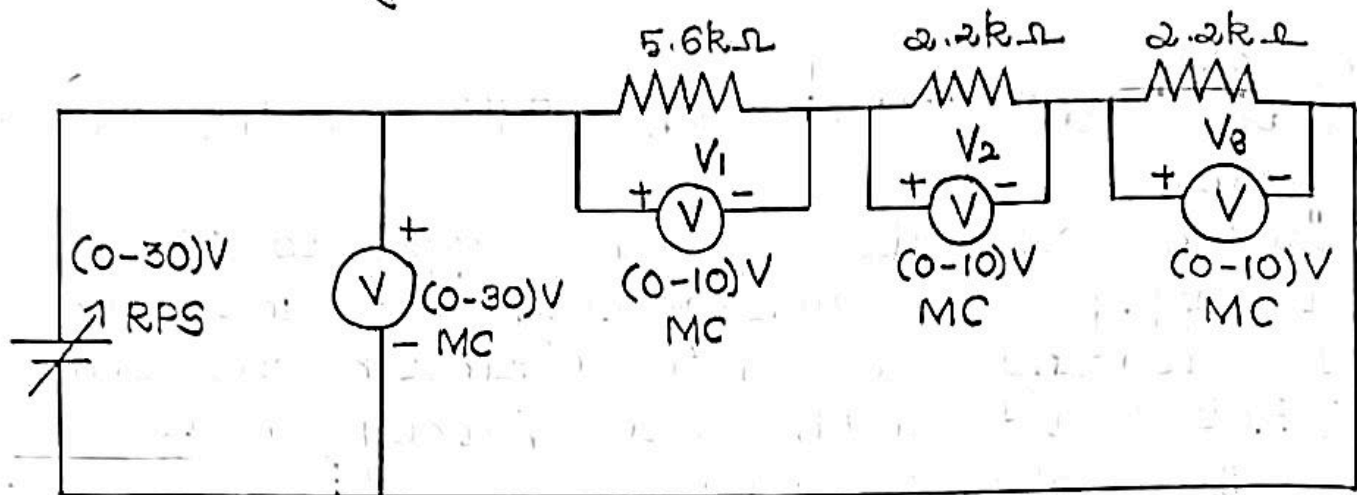
Procedure:

- (i) Make the connections as per the circuit diagram.
- (ii) Switch ON the power supply to RPS and apply a voltage (say 10V) and take the reading of both voltmeter and ammeter.
- (iii) Plot a graph with V along x -axis and I along y -axis.
- (iv) The graph will be a straight line which verifies Ohm's law.

Result:

Thus the Ohm's law is verified for the given circuit.

Circuit diagram: (KVL).



Tabular column:

Sl No.	Supply Voltage	Experimental value				Theoretical value			
		V ₁ (Volts)	V ₂ (Volts)	V ₃ (Volts)	V ₁ +V ₂ +V ₃ (Volts)	V ₁ (Volts)	V ₂ (Volts)	V ₃ (Volts)	V ₁ +V ₂ +V ₃ (Volts)
1	5	2.9	1	1	4.9	2.8	1.1	1.1	5
2	10	5.2	2	2	9.2	5.6	2.2	2.2	10
3	15	8	3	3	14	8.4	3.3	3.3	15
4	20	10.5	4.5	4.5	19.5	11.2	4.4	4.4	20

Model calculation:

(i) Total resistance = $5.6 \times 10^3 + 2.2 \times 10^3 + 2.2 \times 10^3$
 $= 10 \text{ k}\Omega$

Total current, $I = \frac{V}{R} = \frac{5}{10 \times 10^3} = 0.5 \text{ mA}$

$V_1 = IR_1 = 0.5 \times 10^{-3} \times 5.6 \times 10^3 = 2.8 \text{ V}$

$V_2 = IR_2 = 0.5 \times 10^{-3} \times 2.2 \times 10^3 = 1.1 \text{ V}$

$V_3 = IR_3 = 0.5 \times 10^{-3} \times 2.2 \times 10^3 = 1.1 \text{ V}$

Total voltage = $2.8 + 1.1 + 1.1 = 5 \text{ V}$

1) b) Verification of Kirchhoff's laws

Aim: ci) To verify Kirchhoff's voltage law
 cii) To verify Kirchhoff's current law

Apparatus Required:

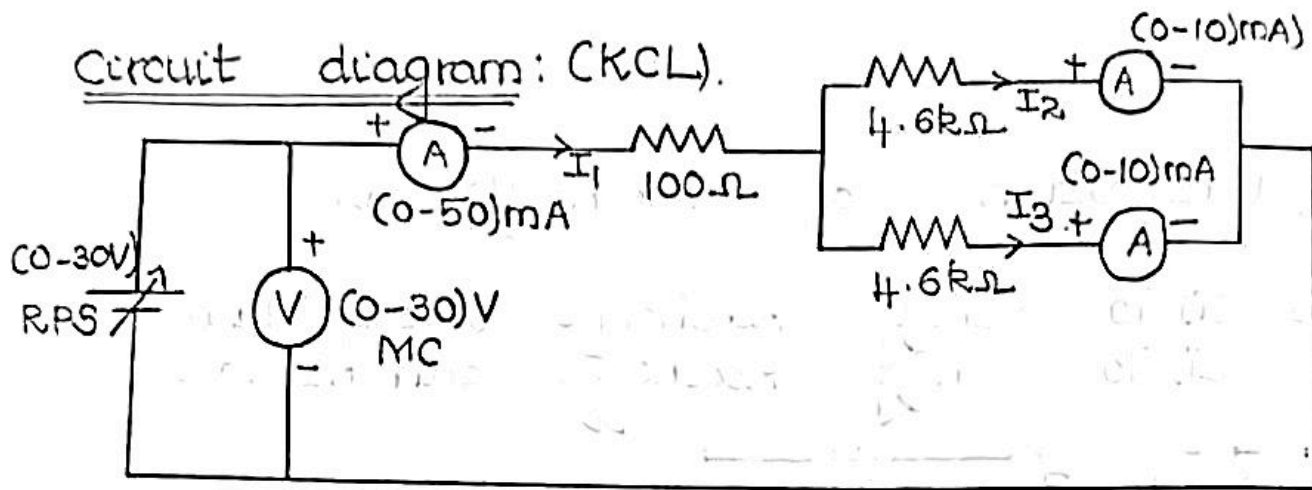
S.No	Name of the equipment	Range	Quantity
1.	RPS (Regulated Power Supply)	(0-30V)	1
2.	Bread board	-	1
3.	Ammeter	(0-50)mA (0-10)mA	1 2
4.	Voltmeter	(0-30)V (0-10)V	1 3
5.	Resistor	5.6k Ω , 2.2k Ω 100 Ω , 4.6k Ω	1, 2 1, 2
6.	Connecting wires	-	As required

Theory:

Kirchhoff's voltage law: (KVL)

Kirchhoff's voltage law states that the algebraic sum of all voltages around any closed loop in an electrical circuit is always equal to zero. This means that the total voltage

Circuit diagram: (KCL).



Tabular column:

Sl. No.	Supply voltage	Experimental value				Theoretical value			
		I_1 (mA)	I_2 (mA)	I_3 (mA)	$I_2 + I_3$ (mA)	I_1 (mA)	I_2 (mA)	I_3 (mA)	$I_2 + I_3$ (mA)
1.	2.7	1	0.57	0.57	1.14	1.125	0.56	0.56	1.12
2.	5.3	2	1.12	1.12	2.24	2.21	1.10	1.10	2.20
3.	8	3	1.67	1.67	3.34	3.33	1.67	1.67	3.34
4.	10.2	4	2.12	2.12	4.24	4.25	2.13	2.13	4.26

Model Calculation:

(i) Supply voltage (V) = 2.7V
 Total Resistance = $100 + \left\{ \frac{4.6 \times 10^3 \times 4.6 \times 10^3}{4.6 \times 10^3 + 4.6 \times 10^3} \right\}$

= $2.4k\Omega$

$I_1 = \frac{V}{R} = \frac{2.7}{2.4k\Omega} = 1.125mA$

$I_2 = I_3 = \frac{1.125 \times 10^{-3} \times 4.6 \times 10^3}{4600 + 4600}$

= $\frac{5.175}{9200}$

= 5.6×10^{-4} (or) 0.56mA

risers provided by sources are exactly balanced by the total voltage drops across the circuit elements in that loop.

Kirchoff's current law: (KCL)

Kirchoff's current law states that the algebraic sum of currents entering and leaving a junction in an electrical circuit is always zero.

KCL is derived from the principle of conservation of charge and is essential for analyzing complex circuits, especially when multiple branches are connected at a single junction.

Procedure:

(i) Connections are given as per the circuit diagram.

(ii) Apply dc voltage to the circuit from the RPS.

(iii) Tabulate the voltmeters and ammeters readings for the corresponding experiment

(iv) Increase the voltage and repeat steps 2 and 3.

(v) Switch off the power supply after bringing RPS to the minimum voltage position.

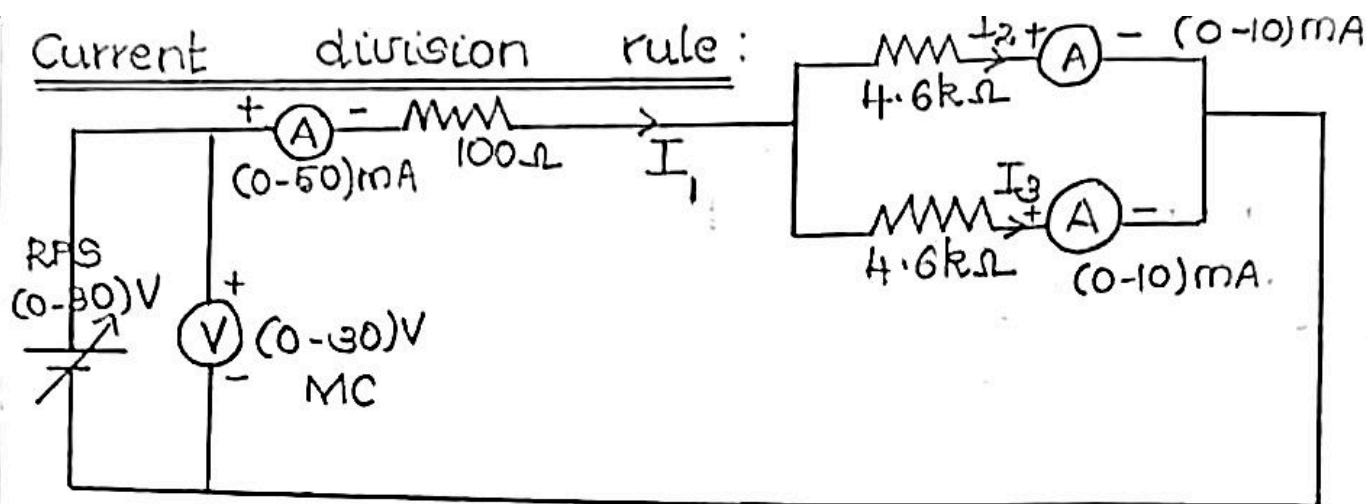
Marks obtained:

Theoretical calculations	20	
Observation	20	
Execution of practical examples	30	
Viva	10	
Record	20	
Total score	100	
Date of experiment		
Date of record submission		Faculty Signature

Result:

Thus the Kirchhoff's voltage and current law is verified for the given circuit.

Current division rule:



Tabular column:

Sl. No.	Supply Voltage	Experimental value				Theoretical value			
		I_1 (mA)	I_2 (mA)	I_3 (mA)	$I_2 + I_3$ (mA)	I_1 (mA)	I_2 (mA)	I_3 (mA)	$I_2 + I_3$ (mA)
1.	2.7	1	0.57	0.57	1.14	1.125	0.56	0.56	1.12
2.	5.3	2	1.12	1.12	2.24	2.21	1.10	1.10	2.20
3.	8	3	1.67	1.67	3.34	3.33	1.67	1.67	3.34
4.	10.2	4	2.12	2.12	4.24	4.25	2.13	2.13	4.26

Model Calculation:

(i) Supply voltage (V) = 2.7V

$$\text{Total Resistance} = 100 + \left\{ \frac{4.6 \times 10^3 \times 4.6 \times 10^3}{4.6 \times 10^3 + 4.6 \times 10^3} \right\}$$

$$= 2.4 \text{ k}\Omega$$

$$I_1 = \frac{V}{R} = \frac{2.7}{2.4 \text{ k}\Omega} = 1.125 \text{ mA}$$

$$I_2 = I_3 = \frac{1.125 \times 10^{-3} \times 4.6 \times 10^3}{4600 + 4600}$$

$$= \frac{5.175}{9200} = 5.6 \times 10^{-4} \text{ (or) } 0.56 \text{ mA}$$

Ex. No. 2 Verification of current and voltage division ruleAim:

To calculate the individual branch currents and total current drawn from the power supply using current and voltage division rule.

Apparatus Required:

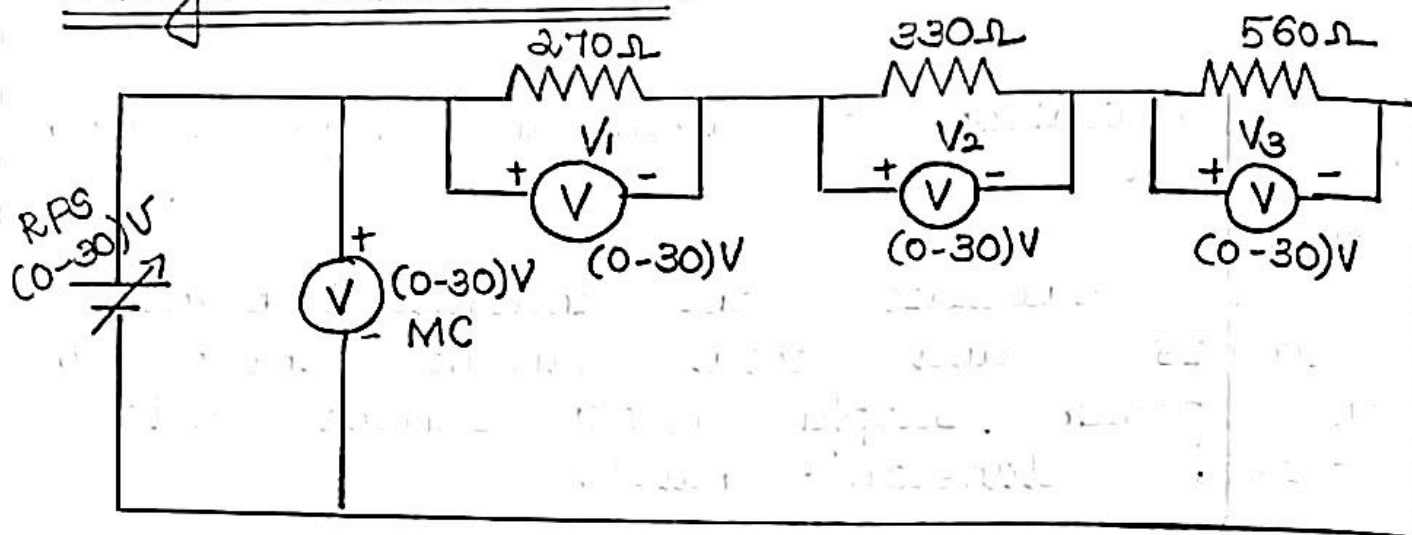
S.No	Apparatus Name	Range	Quantity
1.	Regulated power supply	(0-30)V	1
2.	Ammeter	(0-50)mA	1
		(0-10)mA	2
3.	Voltmeter		
4.	Resistor		
5.	Bread board and connecting wires	-	As required

Theory:

The current division rule and voltage division rule are two fundamental concepts in electrical circuits used to analyze how current and voltage are distributed among components.

The voltage division rule is applied in series circuits. When resistors are connected in series with a voltage

Voltage division rule:



Tabular column:

Sl. No.	RPS Voltage (V)	Experimental value				Theoretical value			
		V ₁	V ₂	V ₃	V ₁ +V ₂ +V ₃	V ₁	V ₂	V ₃	V ₁ +V ₂ +V ₃
1.	5	1	1	2	4	1.16	1.42	2.41	5
2.	10	2	3	4	9	2.32	2.84	4.83	10
3.	15	3	4	7	14	3.49	4.26	7.24	15
4.	20	4	5	10	19	4.65	5.68	9.66	20

Model calculation:

Case (i): Voltage = 5V

$$\text{If } R_1 = 270\Omega; \quad V_1 = \frac{5 \times 270}{1160} = 1.16V$$

$$V_2 = \frac{5 \times 330}{1160} = 1.42V$$

$$V_3 = \frac{5 \times 560}{1160} = 2.41V$$

$$V = V_1 + V_2 + V_3 = 1.16 + 1.42 + 2.41 = 5V$$

source, the total voltage is divided among the resistors in proportion to their resistances.

The current division rule is applied in parallel circuits. When resistors are connected in parallel across a current source, the total current divides among the branches inversely proportional to their resistances.

Procedure:

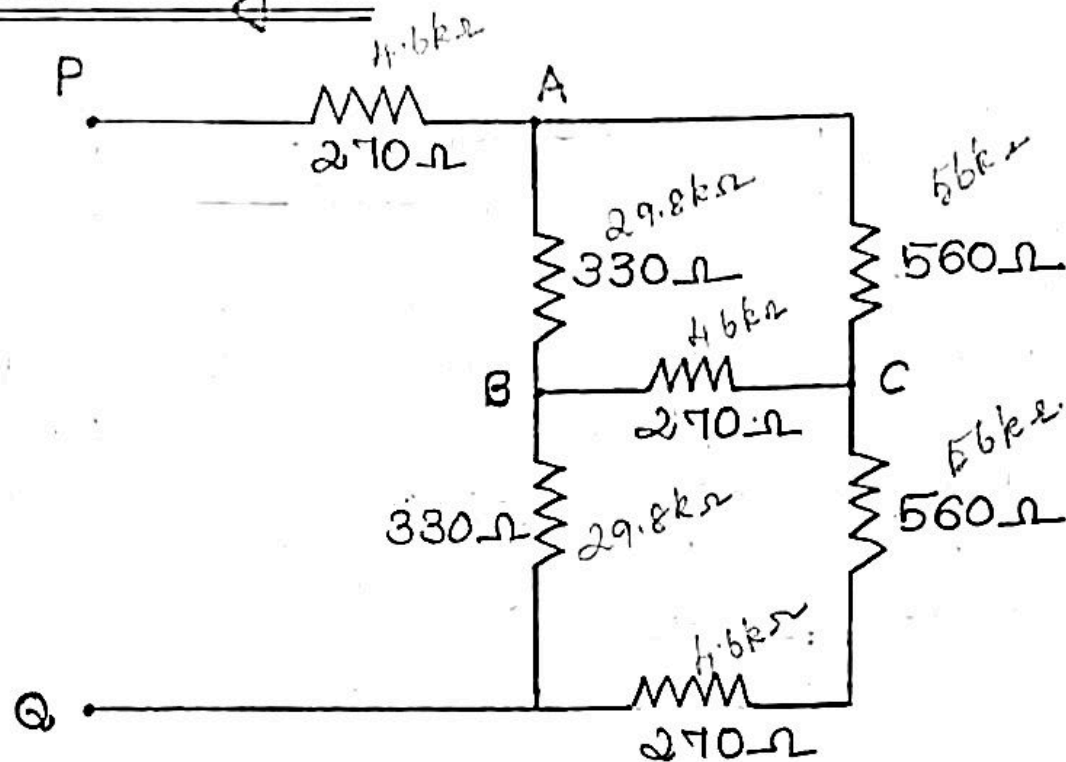
- (i) Give the connections as per the circuit diagram.
- (ii) Set a particular value in RPS.
- (iii) Note down the corresponding ammeter and voltmeter readings.
- (iv) Repeat the same for different voltages and currents.

Marks obtained:

Theoretical calculations	20	
Observation	20	
Execution of practical examples	30	
Viva	10	
Record	20	
Total score	100	
Date of experiment		
Date of Record submission		Faculty signature

Result: Thus the individual branch currents and total current drawn from the power supply are calculated using current and voltage division rules.

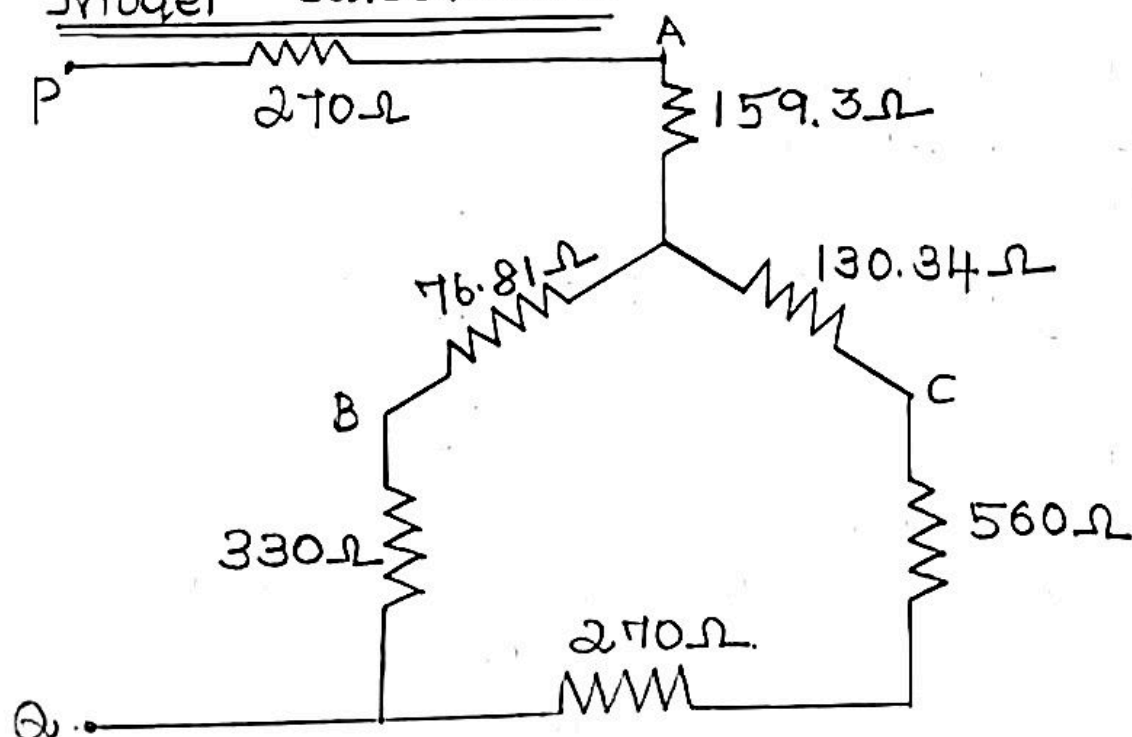
Circuit diagram:



Tabular column:

S.No	Theoretical value (R_{PQ}) in ohm	Measured value (R_{PQ}) in ohm
1.	715.05Ω	716Ω

Model calculation:



Ex. No. 3

Verification of star delta transformation using Resistance Reduction Technique.

Aim:

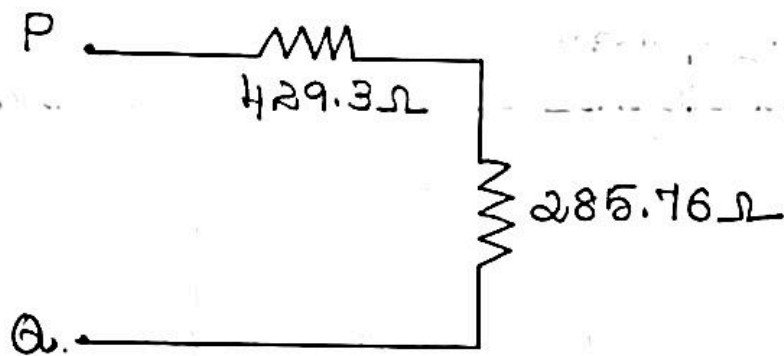
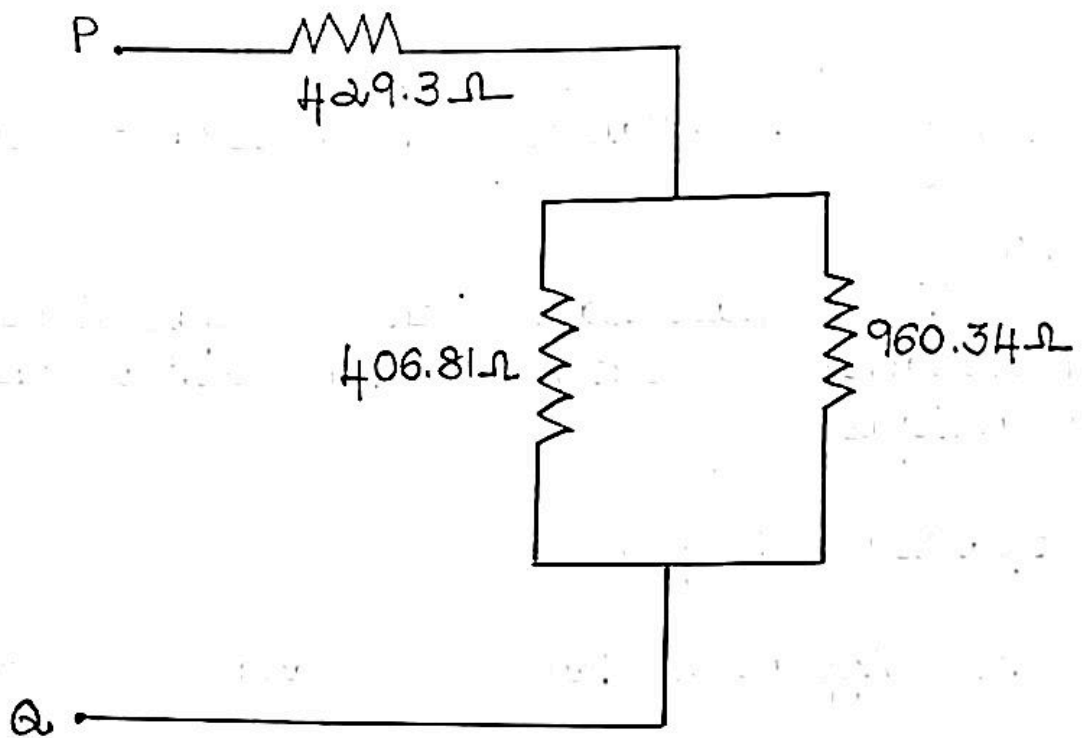
To calculate the equivalent circuit resistance using star delta transformation technique

Apparatus Required:

S.No.	Apparatus Name	Range	Quantity
1.	Resistors	330 Ω , 270 Ω , 560 Ω	2, 3, 2
2.	Bread board		1
3	Connecting Wires		As required

Theory:

The star delta transformation is a mathematical technique used in circuit analysis to simplify complex resistor networks that are neither purely in series nor parallel. In the star configuration, three resistances are connected to a common central node while in delta configuration, the three resistances form a closed triangle. This technique is widely used in circuit analysis, bridge networks, and three phase power systems.



$$= 429.3 + 285.76$$

$$R_{PQ} = 715.05\ \Omega$$

Procedure:

1. Give the connections as per the circuit diagram.
2. Determine the equivalent resistance of the circuit between P and Q using star-delta transformation technique.
3. Verify the same by connecting Multimeter across PQ.

Marks obtained:

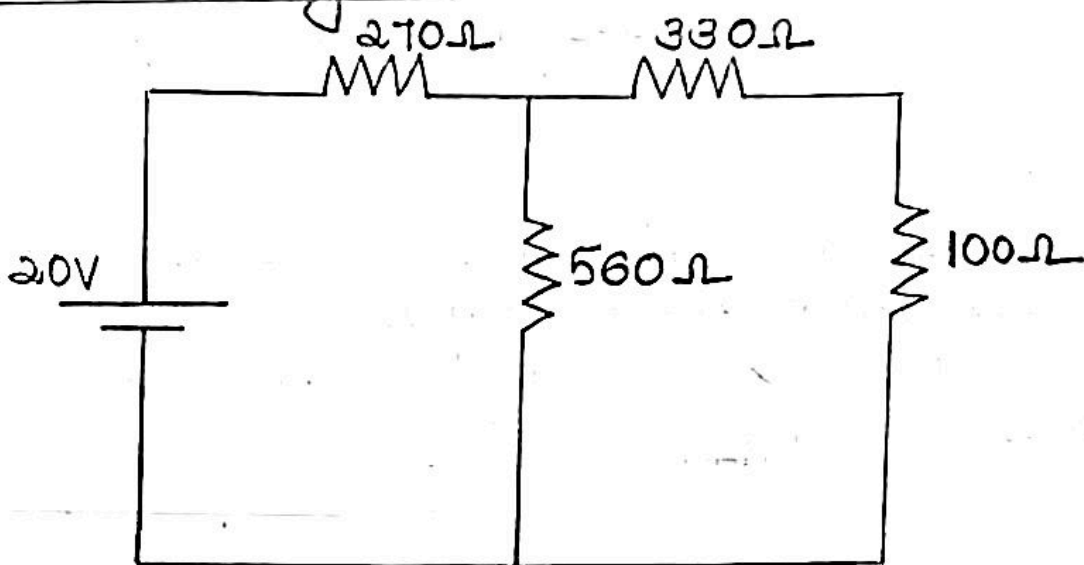
Theoretical calculations	20	
Observation	20	
Execution of practical examples.	30	
Viva	10	
Record	20	
Total score	100	
Date of experiment		
Date of Record submission		Faculty sign

Result:

Thus the equivalent circuit resistance is obtained using star delta transformation technique.

Thevenin's theorem:

Circuit diagram:

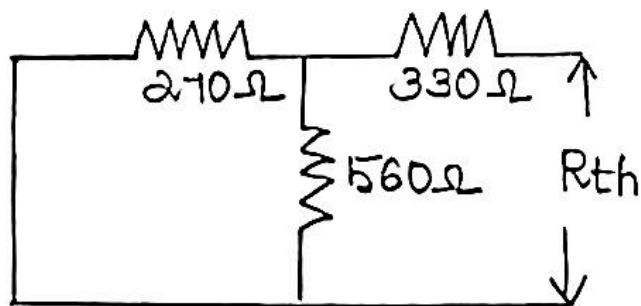


Tabular column:

Sl. No.	Supply voltage (V)	Theoretical values			Measured values		
		$R_{th}(\Omega)$	$V_{th}(V)$	$I_L(A)$	$R_{th}(\Omega)$	$V_{th}(V)$	$I_L(mA)$
1.	20	512.16	13.49	0.022	512	14	22

Model calculation:

To find R_{th} :



$$R_{th} = 330 + \frac{270 \times 560}{270 + 560}$$
$$= 182.16\Omega + 330\Omega$$

$$R_{th} = 512.168\Omega$$

Ex. No: 4

Verification of Thevenin's and Norton's theoremAim:

To verify the equivalent circuit parameters of Thevenin's and Norton's theorems theoretically and practically.

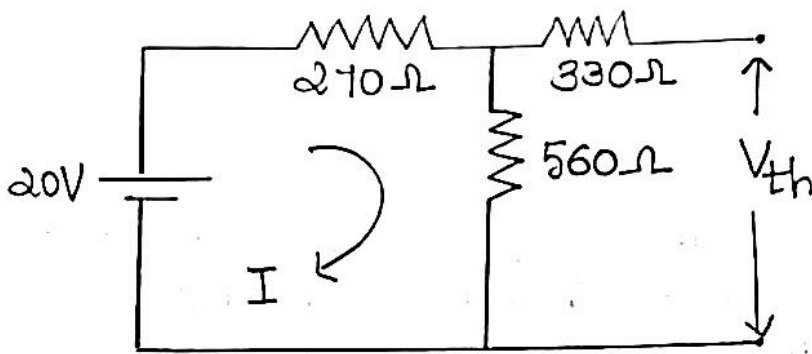
Apparatus Required:

S.No	Apparatus Name	Range	Quantity
1.	RPS (Regulated Power supply)	(0-30)V	1
2.	Voltmeter	(0-100)V	2
3.	Ammeter	(0-100)mA	2
4.	Resistor	270 Ω , 330 Ω , 560 Ω	1
5.	Multimeter	-	1
6.	Bread board & Connecting wires	-	As required

Theory:

Thevenin's and Norton's theorems are fundamental tools in circuit analysis that simplify complex electrical networks into easier equivalents. Thevenin's theorem states that any linear two terminal circuit can be replaced by a single voltage source in series with a resistor. Norton's theorem states that the same circuit

To find V_{th} :

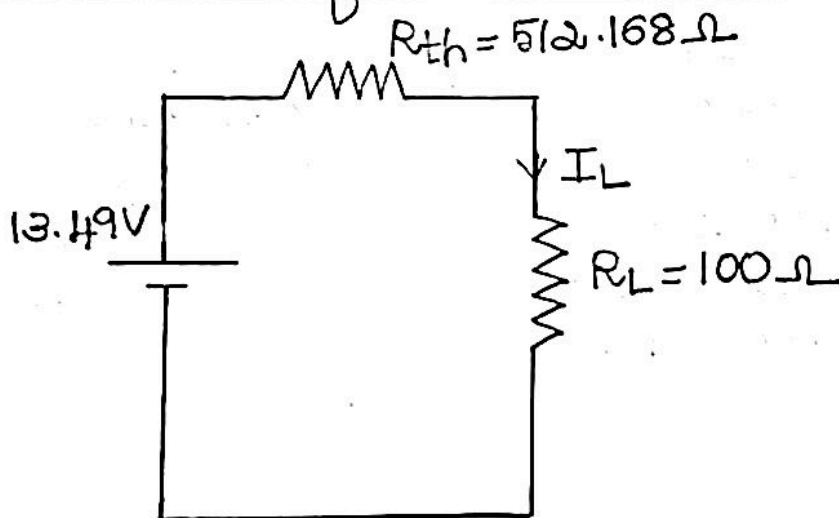


$$I = \frac{V}{R} = \frac{20}{270 + 560} = 0.024 \text{ A}$$

$$V_{th} = V_{560\Omega} = 0.024 \times 560$$

$$\boxed{V_{th} = 13.49 \text{ V}}$$

To find I_L from Thevenin's equivalent circuit



$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

$$= \frac{13.49}{512.17 + 100} = 0.022 \text{ A}$$

can also be represented as a current source in parallel with a resistor.

Procedure: (i) Thevenin's theorem:

(i) Give the connections as per the circuit diagram.

(ii) Measure R_{th} by using a multimeter by open circuiting the current source; short circuiting the voltage source and open circuit R_L .

(iii) Measure V_{th} across A and B. (Open circuit R_L)

(iv) Measure load current I_L through R_L

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

(v) Draw Thevenin's equivalent circuit.

(ii) Norton's theorem:

(i) Give the connections as per the circuit diagram.

(ii) Measure R_{th} using a multimeter (O.C. the current source, S.C. the voltage source and open circuit R_L)

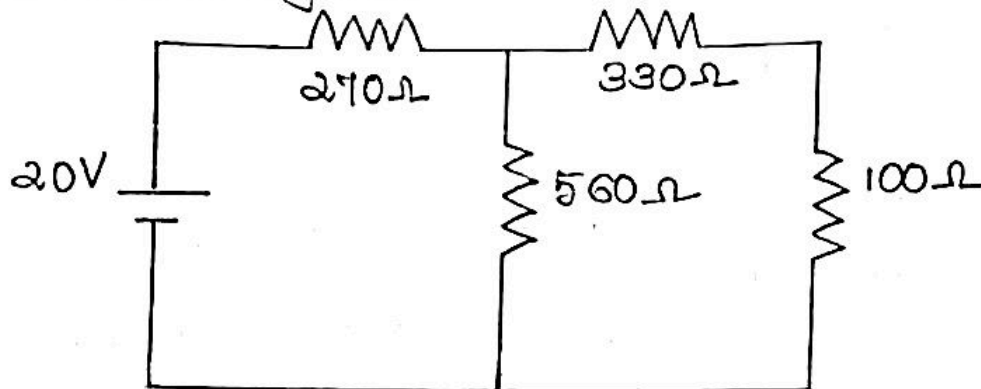
(iii) Measure I_N through A and B (short circuit R_L)

(iv) Measure load current I_L through R_L

(v) Draw Norton's equivalent circuit.

Norton's theorem:

Circuit diagram:

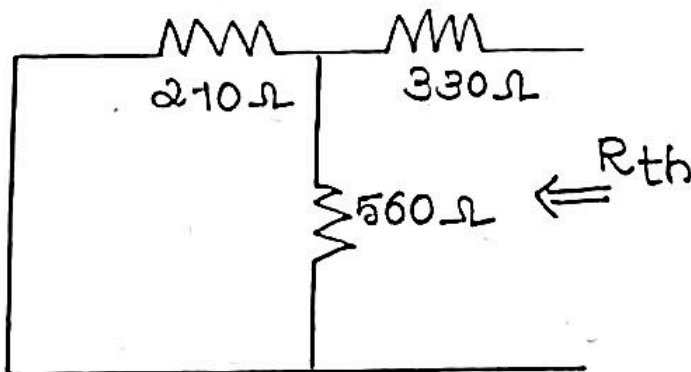


Tabular column:

Sl. No.	Supply Voltage (V)	Theoretical values			Measured values		
		R_{th} (Ω)	I_{sc} (A)	I_L (A)	R_{th} (Ω)	I_{sc} (mA)	I_L (mA)
1.	20	512.17	0.026	0.022	512	25	22

Model Calculation:

To find R_{th} :

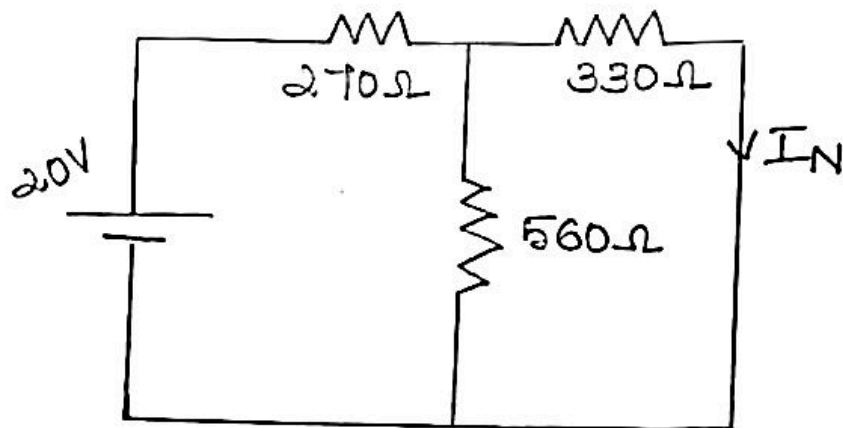


$$R_{th} = 330 + \left\{ \frac{270 \times 560}{270 + 560} \right\}$$

$$= 182.16 + 330$$

$$= 512.17 \Omega$$

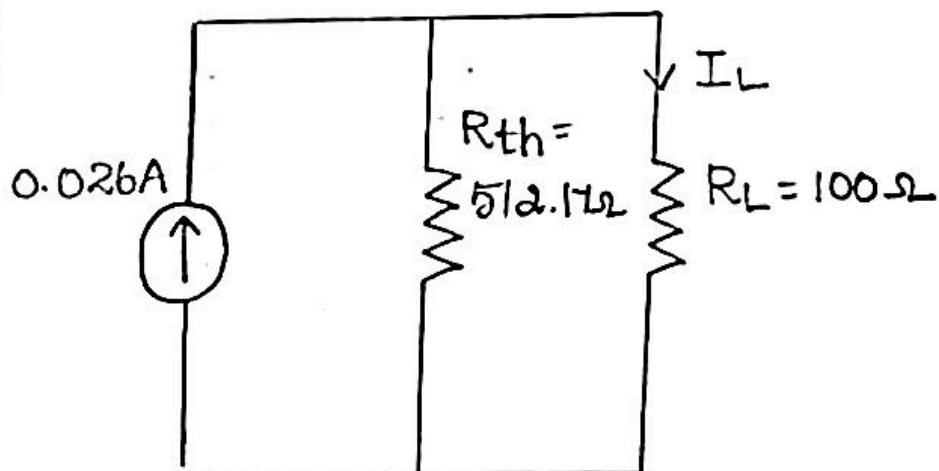
To find I_{SC} (or) I_N :



$$I_T = \frac{V}{R_{eq}} = \frac{20}{207.64 + 270} = 0.0418A$$

$$I_N \text{ (or) } I_{SC} = \frac{I_T \times 560}{560 + 330}$$
$$= 0.026A$$

Norton's equivalent circuit:



$$I_L = \frac{0.026 \times 512.17}{512.17 + 100} = 0.022A$$

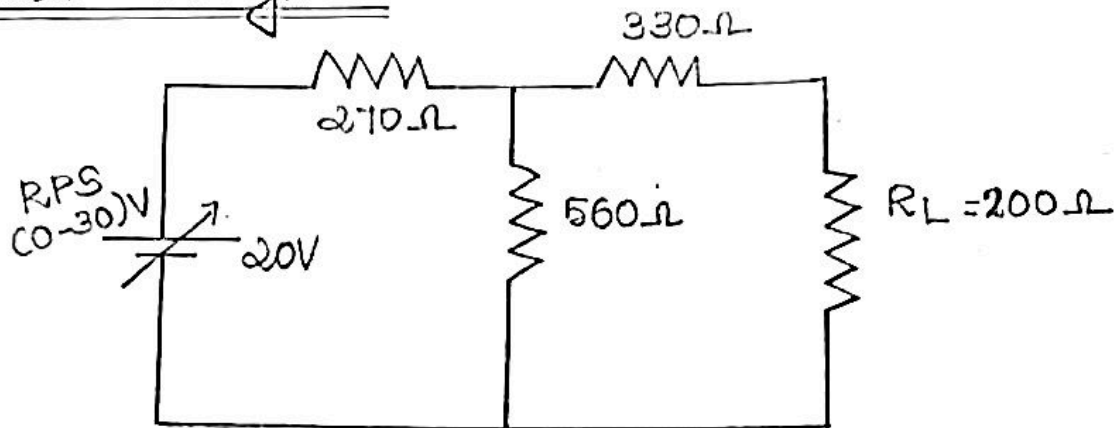
Marks obtained:

Theoretical calculations	20	
Observation	20	
Execution of practical examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		
Date of record submission		Faculty sign

Result:

Thus the equivalent circuit parameters are obtained using Thevenin's and Norton's theorem.

Circuit diagram:



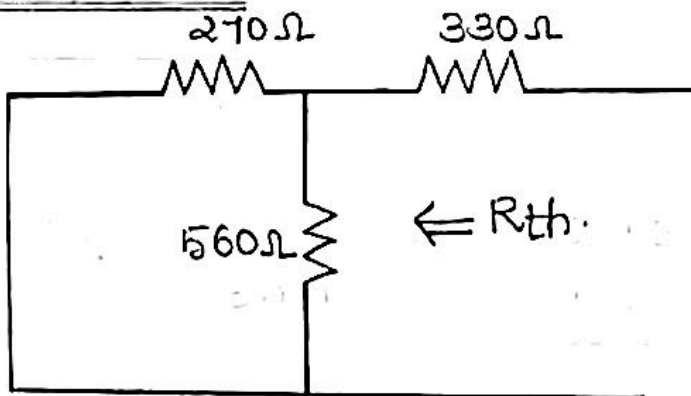
Tabular column:

$R_{th} = \underline{\hspace{2cm}} \Omega$

Sl. No.	$R_L (\Omega)$	$I_L (mA)$	Power = $I_L^2 R_L$ (mW)
1.	200	18.6	69.2
2.	560	12.3	84.7
3.	1000	9	81

Model calculation:

To find R_{th}



$$R_{th} = 330 + \left\{ \frac{270 \times 560}{270 + 560} \right\}$$

$$R_{th} = 512.168 \Omega$$

Ex.No:5 Verification of maximum power transfer theorem and superposition theorem

a) Verification of Maximum Power Transfer theorem

Aim:

To verify Maximum Power Transfer Theorem.

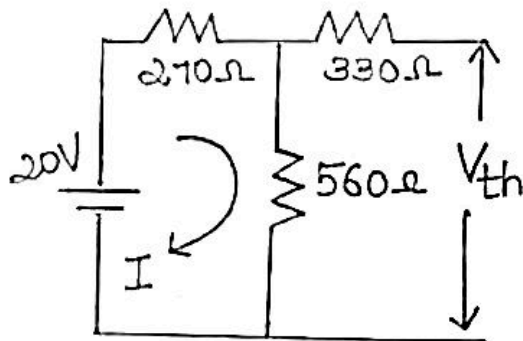
Apparatus Required:

Sl.No.	Apparatus Name	Range	Quantity
1.	R.P.S	(0-30)V	1+1
2.	Voltmeter		
3.	Ammeter		
4.	Resistor		
5.	Multimeter	-	1
6.	Bread board and connecting wires	-	Required

Theory:

The maximum power transfer theorem states that maximum power is delivered to a load when the load resistance equals the Thevenin resistance of the source network. It is mainly applied in communication systems and matching circuits to ensure efficient power transfer. On the other hand, the superposition

To find V_{th} :



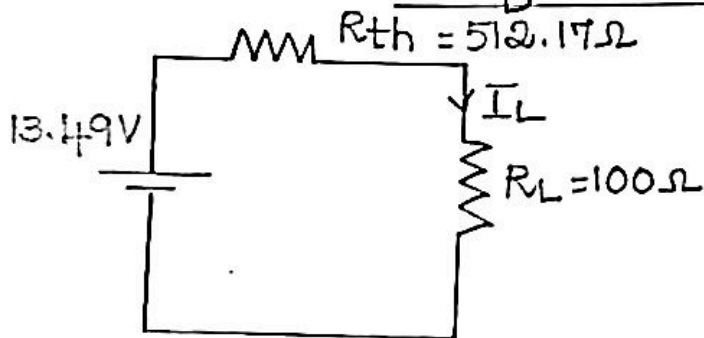
$$I = V/R$$

$$= \frac{20}{270 + 560} = 0.024 \text{ A}$$

$$V_{th} = V_{560\Omega} = 0.024 \times 560$$

$$V_{th} = 13.49 \text{ V}$$

To find I_L from Thevenin's equivalent circuit:



$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

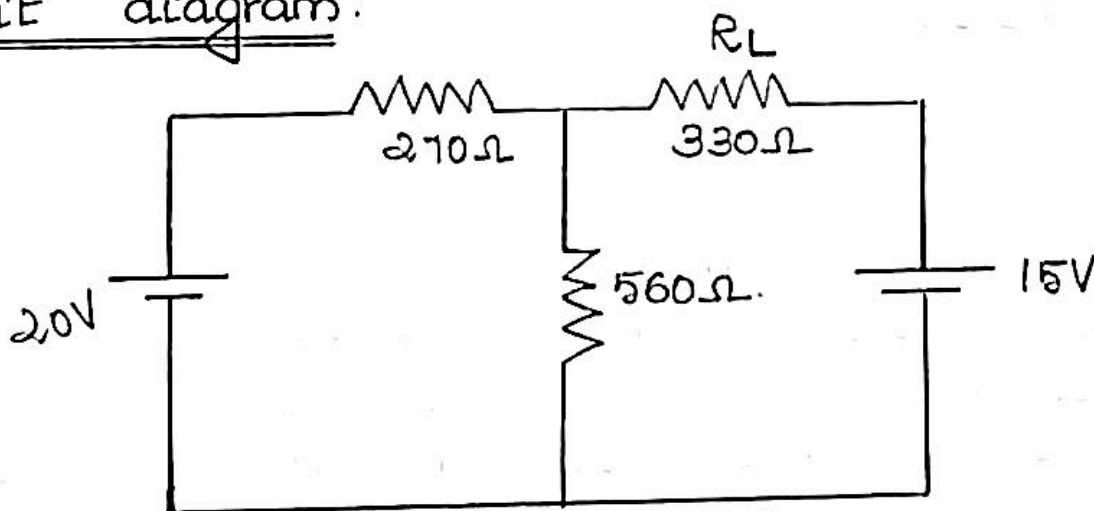
$$= \frac{13.49}{512.17 + 100} = 0.022 \text{ A}$$

To find P_{max} :

$$P_{max} = \frac{V_{th}^2}{4 R_{th}} = \frac{13.49^2}{4 \times 512.168} = \frac{181.98}{2048.672} = 0.0888 \text{ W}$$

Superposition theorem:

circuit diagram:



theorem is used in linear circuits with multiple independent sources, stating that the total current or voltage in any element is the algebraic sum of the effects produced by each source acting alone, with other sources replaced by their internal resistances. While the maximum power transfer theorem deals with efficiency of power delivery, the superposition theorem simplifies circuit analysis involving multiple sources.

Procedure:

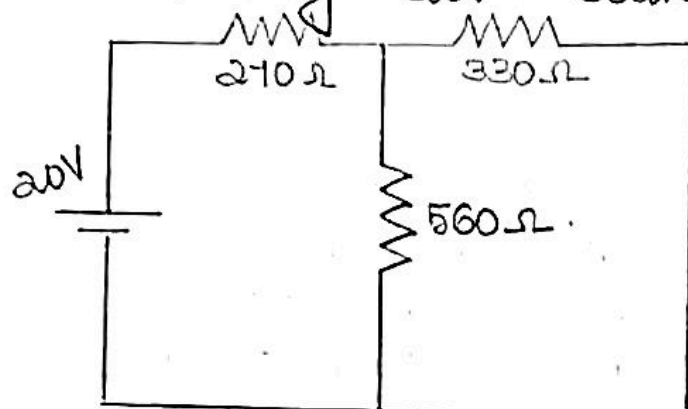
(i) Maximum power theorem:

1. Connections are given as per the circuit diagram
2. Measure R_{th} using a multimeter
3. Measure V_{th} across 560Ω
4. Measure load current I_L through R_L
5. Calculate the maximum power transferred to the load.

(ii) Superposition theorem:

1. Give the connections as per the circuit diagram
2. Measure current flow through 330Ω by connecting both the supplies
3. Short circuit $15V$
4. Measure current flow through 330Ω by connecting $20V$ supply
5. Short circuit $20V$
6. Measure current flow through 330Ω by

Case (i): Only 20V source is considered



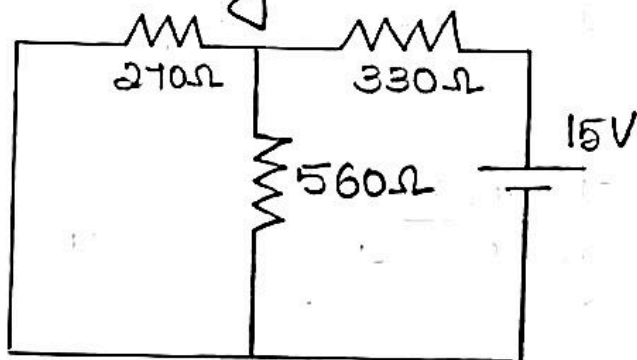
$$I_T = \frac{20}{270 + \left\{ \frac{330 \times 560}{330 + 560} \right\}}$$

$$= \frac{20}{270 + 207.640}$$

$$I_T = 0.0418 \text{ A}$$

$$I_{330\Omega}^I = \frac{I_T \times 560}{560 + 330} = \frac{0.0418 \times 560}{890} = 0.0263 \text{ A}$$

Case (ii): Only 15V source is considered:



$$I_T = \frac{15}{330 + \left\{ \frac{270 \times 560}{270 + 560} \right\}}$$

$$= \frac{15}{330 + 182.168}$$

$$I_T = \frac{15}{512.168} = 0.029 \text{ A}$$

$$I_{330\Omega}^{II} = 0.029 \text{ A}$$

$$I_{330\Omega} = I_{330\Omega}^{II} - I_{330\Omega}^I$$

$$= 0.029 - 0.0263$$

$$I_{330\Omega} = 0.0027 \text{ A}$$

Tabular column:

Sl. No.	Measured current when both supplies are given	Calculated current thro' 330Ω		Measured current thro' 330Ω		Net current
		V = 20V	V = 15V	V = 20V	V = 50V	
1	2.7 mA	26 mA	29 mA	26 mA	28 mA	2 mA

connecting 15V supply.
 7. Verify the net current flowing through
 330 Ω resistor.

Marks obtained

Theoretical calculations	20	
Observation	20	
Execution of practical examples	30	
Viva	10	
Record	20	
Total score	100	
Date of experiment		
Date of record submission		
		Faculty Sign

Result:

Thus the maximum power transfer theorem and superposition theorem are verified.